



Bidding behavior in experimental auctions with positive and negative values



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HIGHLIGHTS

- We conduct experimental auctions allowing negative bids for an exchange of one good for another.
- Results suggest that positive WTP bids are demand revealing but subjects overbid their negative WTA values.
- Overbidding of negative values was more pronounced in a random n th price auction than in a 5th price auction.

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ABSTRACT

We test for consistency between bids from endowment auctions that allow positive or negative bids for an exchange between two items, and bids from prior full bidding auctions for the same items. Results suggest that positive WTP bids in the endowment auction are demand revealing but subjects overbid (in absolute value) their negative WTA values. WTA bids were lower in a random n th price auction than in a 5th price auction.

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1. Introduction

Goods such as irradiated or genetically modified foods are valued positively by some consumers and negatively by others. Mixed logit estimation with choice experiment data readily accommodates simultaneous estimation of positive and negative values, but the same is not true for valuation experiments. Experimental auctions typically elicit either willingness-to-pay or willingness-to-accept, but rarely allow negative bids (Lusk and Shogren, 2007). Instead, when faced with negative valuations, researchers typically truncate bids at zero, or separate participants based on preference and elicit WTP to upgrade to the preferred alternative (Fox et al., 1998).

Reluctance to allow negative bids may stem from concerns about comprehension of the auction mechanism but also from concerns about whether subjects would bid strategically to enhance earnings (Dickinson and Bailey, 2002). Parkhurst et al.

(2004) reported instances of extreme underbidding of negative values in an induced value experiment, but in general, subjects with negative values actually tended to overbid (not reveal the full extent of the negative valuation) in a 2nd price auction. In the same study, results from a random n th price auction were less precise but did not reveal the tendency to overbid negative values.

We investigate bidding behavior when negative bids are allowed for privately valued goods. We focus on whether subjects with negative values tend to bid strategically, and the performance of random n th and 5th price auctions when negative bids are allowed. Any tendency to either under- or over-bid a negative value would be important to recognize should practitioners begin to routinely allow negative bids in experiments. Our results suggest that subjects tend to underbid WTA values, and negative bids from random n th auctions tend to be below those from 5th price auctions.

2. Experimental design

Participants were students at Kansas State University. For participating they were paid \$5 and received a pizza meal. Five sessions were held, each with ten participants. Two sessions used

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a 5th price auction, two used a random n th price auction, and one session used both auctions.

On arrival, participants signed a consent form, received the \$5 payment, were provided an ID number, and completed a questionnaire.¹ In Stage 1, participants examined two products – a 10 oz package of Trail Mix (TM) and a 6 oz package of Dried Fruit (DF). They were asked to indicate a preference – i.e., I prefer Trail Mix; I prefer Dried Fruit; I like them both equally. Participants were told that they would participate in 2nd price auctions for both products. A practice auction using a candy bar was used for demonstration, and the cash payment by the winning bidder emphasized that the exercise was not hypothetical. Participants then bid for TM and DF, but were informed that only one of the auctions, selected by coin toss, would be binding. The binding auction and winning bidder would be determined at the conclusion of the experiment, after bidding in the subsequent Stage 2 auction was completed.

Stage 2 featured endowment auctions (Corrigan and Rousu, 2006) in which participants bid either positive WTP or negative WTA to exchange one good for another. Again, only one auction (the number of auctions was not revealed) would be binding, and there would be $N - 1$ winning bidders each paying an amount equal to the N th bid. In sessions using the 5th price auction, $N = 5$ was revealed prior to bidding. In sessions using a random n th auction, participants were told that N would be a randomly drawn number between 2 and 8, determined after the bidding. An illustration showed how the auction worked in a scenario with seven subjects bidding to exchange an oven mitt for a coffee mug. In the illustration, three people preferred the coffee mug (and bid positive WTP for the exchange), one was indifferent (bid zero), and three preferred the oven mitt (bid negative WTA for the exchange). Positive, zero, and negative bids were illustrated on a line. With $N = 5$, the result was 4 bidders making the exchange and getting paid the amount of the (negative) 5th highest bid. To ensure that subjects understood the mechanism, they were asked a series of questions – what would happen if $N = 2$, etc. To further familiarize them with the procedure, another practice auction was conducted using candy bars.

We then conducted Auction A, in which participants were endowed with DF and could bid to exchange it for TM. On a bid sheet, participants were asked to write “the most I am willing to pay” or “the minimum I would accept” to make the exchange, and were instructed to also mark their bid along a line. The monitor checked sheets for consistency between written bids and the marked bid on the line and sought clarification in the event of a discrepancy. After bids were collected, participants handed back the DF and proceeded to Auction B in which the endowment was reversed and subjects bid to exchange TM for DF.²

In our analysis, performance of the endowment auctions is assessed in terms of consistency between the initial bid differential between the two products in stage 1, and bids for upgrades or downgrades in stage 2.

3. Results and discussion

After checking initial bids and stated preferences, bids from three subjects (all from the session using both auction types) were discarded for inconsistency, i.e., stating a preference for good A, but submitting a larger bid for good B. Table 1 summarizes bids for the remaining 27 observations for each auction.

Table 1
Stage 1 and Stage 2 bids.

Auction type		Stage 1		Stage 2	
		Bid for TM	Bid for DF	DF → TM	TM → DF
5th price ($N = 27$)	Mean	1.58	1.47	0.03	−0.54
	St. Dev.	1.04	1.19	1.09	1.70
Random n th ($N = 27$)	Mean	1.74	1.62	−0.35	−0.28
	St. Dev.	1.13	1.33	1.65	2.00

Stage 1 bids are slightly higher for TM than for DF but the difference is not significant ($p = 0.43$).³ Stage 1 bids suggest that the average bid for the DF → TM exchange in Stage 2 should be \$0.11 and \$0.12 in the 5th and random n th auctions. The observed DF → TM averages are lower, insignificantly in the 5th price (\$0.03, $p = 0.58$) but significantly lower in the random n th (−\$0.35, $p = 0.01$). For the opposite TM → DF exchange, average bids in the 5th and random n th auctions (−\$0.54 and −\$0.28) are again below the expected −\$0.11 and −\$0.12, but the differences are not significant ($p = 0.19$, $p = 0.49$). The average bid for the DF → TM exchange is higher in the 5th price auction, while that for the TM → DF exchange is higher in the random n th auction. In Table 2, we re-categorize Stage 2 bids as either WTP or WTA values according to the preference revealed in Stage 1.⁴ “Upgrade WTP” represents bids to exchange to the preferred alternative (whether DF or TM), and, similarly, “Downgrade WTA” represents negative bids to obtain a less preferred alternative. Table 2 values suggest that upgrade WTP is similar to the Stage 1 premium, while downgrade WTA values exceed (in absolute value) the Stage 1 premium. Among 53 pairs of DF → TM, TM → DF bids, we found 16 that could be classified⁵ as exhibiting $WTA \gg WTP$, but only 3 where $WTP \gg WTA$. We test the hypothesis of demand revealing bidding behavior in Stage 2 using the following structure:

$$WTP_i(WTA_i) = \alpha + \beta * PREM_i + \varepsilon_i \quad (1)$$

where $WTP_i(WTA_i)$ is the Stage 2 bid to upgrade (downgrade), and $PREM_i$ is the premium for the preferred good from Stage 1. Stage 2 bids are demand revealing if $\alpha = 0$ and $\beta = 1$ for WTP, and $\alpha = 0$ and $\beta = -1$ for WTA. The estimated equations for each auction, with standard errors in parentheses, are:

5th Price:

$$WTP_i = 0.19 + 0.81 * PREM_i \quad (W = 0.64; p = 0.53) \\ (0.17) \quad (0.18)$$

$$WTA_i = -0.75 - 0.73 * PREM_i \quad (W = 3.73; p = 0.038) \\ (0.31) \quad (0.32)$$

Random N:

$$WTP_i = 0.11 + 0.91 * PREM_i \quad (W = 0.08; p = 0.92) \\ (0.29) \quad (0.24)$$

$$WTA_i = -0.36 - 1.35 * PREM_i \quad (W = 5.98; p = 0.008) \\ (0.33) \quad (0.28)$$

The Wald test values (W) test the joint hypothesis $\alpha = 0$ and $\beta = 1/-1$. The hypothesis of demand revealing bidding is rejected for the downgrade WTA bids in both auctions, but is not rejected for the WTP bids. WTA coefficients indicate that participants demand more compensation to surrender the preferred good than they are willing to pay to obtain it – a result consistent with several experimental studies (Isoni et al., 2011).

³ All p -values show paired t -test results.

⁴ One subject indicated equal preference for TM and DF, bid equal amounts for both products, and bid zero WTP and WTA amounts in Stage 2. Because the Stage 2 bids could not be categorized as either WTP or WTA, the individual's bids were dropped from this analysis.

⁵ Classifying $WTA \gg WTP$ if the absolute value of the negative bid is at least twice the value of the positive bid, or is at least \$1 more than the positive bid.

¹ Instructions are available from the corresponding author.

² Two additional rounds, C and D, were conducted in the session that used both auction types.

Table 2
Upgrade WTP and downgrade WTA bids.

Auction type		Stage 1 premium	Upgrade WTP	Downgrade WTA
5th price (<i>N</i> = 27)	Mean	0.69	0.74	−1.25
	St. Dev.	0.71	0.85	1.22
Random <i>n</i> th (<i>N</i> = 26)	Mean	0.93	0.95	−1.61
	St. Dev.	0.77	1.16	1.49

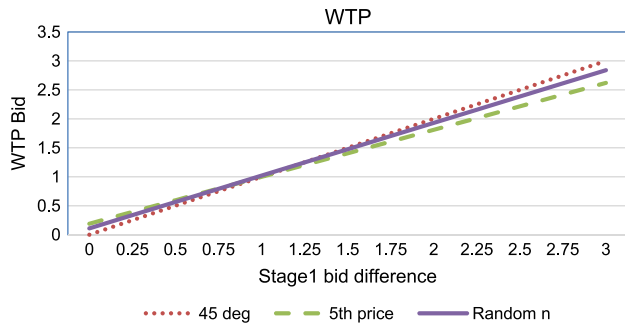


Fig. 1. WTP bids vs Stage 1 premium.

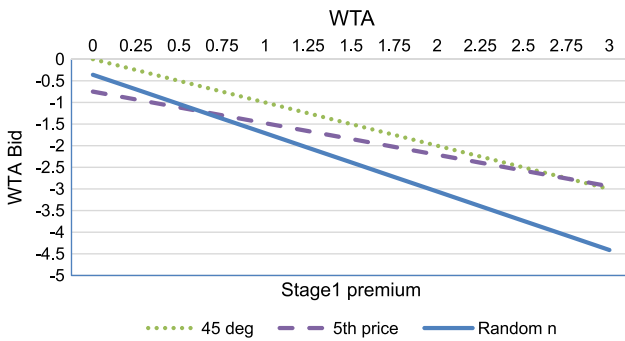


Fig. 2. WTA bids vs Stage 1 premium.

The final objective is to compare performance of the 5th price and random *n*th price auctions. For subjects with large positive or negative values, Parkhurst et al. (2004) found that subjects in a 2nd price auction tended to overbid while bids in the random *n*th auction were unbiased. We investigate whether bidding is similar across auctions using:

$$WTP_i(WTA_i) = \alpha_1 + \beta_1 * PREM_i + \alpha_2 * RandomN + \beta_2 * PREM_i * RandomN + \varepsilon_i \quad (2)$$

where *RandomN* is a dummy variable for the random *n*th auction. Bidding behavior is similar if α_2 and β_2 are both zero. The estimated equations are:

$$WTP_i = 0.19 + 0.81 * PREM_i - 0.07 * RandomN + 0.09 * PREM_i * RandomN \quad (0.21) \quad (0.22) \quad (0.33) \quad (0.31)$$

(*W* = 0.05; *p* = 0.95)

$$WTA_i = -0.75 - 0.73 * PREM_i + 0.39 * RandomN - 0.63 * PREM_i * RandomN \quad (0.30) \quad (0.31) \quad (0.46) \quad (0.42)$$

(*W* = 1.16; *p* = 0.32)

Wald tests fail to reject similar bidding behavior in the two auction mechanisms for both positive (WTP) and negative (WTA) values. However, while the estimated α_2 and β_2 coefficients are both close to zero in the WTP equation, they are of considerably greater magnitude in the WTA equation. Figs. 1 and 2 plot WTP and WTA

bids against the Stage 1 premium. Fig. 2 shows that WTA bids elicited with the random *n*th auction are everywhere below the Stage 1 premium, and that the divergence increases with the Stage 1 premium.

4. Conclusions

Valuation studies often deal with goods or attributes for which some subjects hold positive value, others negative. Auctions permitting simultaneous elicitation of positive and negative bids have advantages over other approaches that truncate bidding at zero, or separate participants based on preferences. Negative bids allow for complete revelation of demand, but may introduce incentives for strategic bidding.

Using an approach that compares values elicited in an endowment auction that permits negative bids with values from prior “full-bidding” auctions for the same two private goods, we show that allowing negative bids did not compromise accurate revelation of positive WTP values in 5th price or random *n*th auctions. We find that WTA exceeds WTP for opposite exchanges of goods, a result consistent with other experimental studies, and we reject the hypothesis that negative WTA bids reveal demand. Random *n*th and 5th price auctions performed similarly in eliciting WTP values. With negative values, the tendency to overbid WTA was more pronounced in the random *n*th auction, a finding similar to Lusk et al. (2004).

We find no reason to discourage practitioners from allowing simultaneous positive/negative bidding in experiments, but our results suggest that negative bids will reflect the common WTP/WTA disparity, and that it may be more pronounced in a random *n*th auction.

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